

A Frequency Dial for Structure-Conditioned Diffusion

SUBMISSION ID

Abstract

A structure-conditioned diffusion adapter is trained for one operating point. Canny edges, a depth map, a sketch: each fixes how tightly the output must follow the condition, and moving that trade-off means training again. We train a single adapter on conditions whose information content is randomised — a high-pass filter whose cutoff r is drawn per sample — and read r back at test time as a dial. On FFHQ-512, structure consistency runs from 0.7469 at $r=0.05$ to 0.2347 at $r=0.3$ while sample diversity rises from 0.3126 to 0.3794, and FID stays between 51.58 and 54.68 over that span. We report the sweep out to $r=1$ but claim the dial only on $r \in [0, 0.3]$, above which the condition retains too little energy for structure consistency to be interpretable. Turning the adapter’s strength down is not the same knob. At matched structure consistency — scale 0.50 reaches 0.2146 and the cutoff $r=0.3$ reaches 0.2347 — the strength setting costs FID 76.46 where the cutoff costs 52.77, so the cutoff dominates on both axes at once.

Two findings came out of building it. The frozen SD VAE destroys a high-pass condition before any adapter can read it: only 0.5773 of the condition survives an encode–decode round trip at $r=0.3$ and 0.0700 at $r=0.7$. And between these two injection designs placement matters more than width — skip-only retains SC 0.5370 against 0.2112 for attention-only, with both paths at 0.6991 — while quadrupling the adapter’s width inside the attention-only design buys far less. The adapter trains 6,611,469 parameters, $55\times$ fewer than ControlNet.

CCS Concepts

• **Computing methodologies** → **Image manipulation**; **Neural networks**;

1. Introduction

ControlNet [ZRA23] and its successors [MWX*23, ZCC*23, QZY*23, LYK*24] solved conditional control and then froze it. An adapter trained on Canny edges follows Canny edges; one trained on depth follows depth. The strength of that following is a property of the trained weights, so an artist who wants the same sketch honoured more loosely has to pick a different checkpoint or accept whatever the one they have was trained to do.

We change what the condition is rather than how hard the model listens to it. The condition is the image after a radial high-pass filter with normalised cutoff r . At $r=0$ the model receives the grayscale image and the task is colourisation. As r grows the filter removes more of the spectrum, until at $r=1$ almost nothing is left: measured on FFHQ, a cutoff of 0.05 already discards 97.5 % of the spectral energy. Training draws r per sample, so one set of weights sees the whole range, and r becomes an inference-time parameter.

Our contributions are:

- A control dial. One adapter spans structure consistency 0.7469 to 0.2347 on $r \in [0.05, 0.3]$ with FID between 51.58 and 54.68 (Sec. 4.2).
- Evidence that the dial is not a rescaled version of adapter strength. At the closest matched pair the two settings reach structure consistency 0.2146 and 0.2347, at FID 76.46 and 52.77 (Sec. 4.3).

- A measurement of how much of a high-pass condition survives the frozen VAE, which explains why latent-space conditioning fails for sparse structure (Sec. 4.4).
- A controlled comparison of injection sites showing that placement matters more than adapter capacity or receptive field (Sec. 4.5).

2. Related work

Conditional control. Text-to-image diffusion [HJA20, SME20, RBL*21, NDR*21, SCS*22] became spatially controllable with ControlNet [ZRA23], which duplicates the encoder and injects residuals through zero convolutions. T2I-Adapter [MWX*23] does the same job with a small feature extractor, IP-Adapter [YZL*23] adds decoupled cross-attention for image prompts, and GLIGEN [LLW*23] grounds generation on boxes. Uni-ControlNet [ZCC*23] and UniControl [QZY*23] fold many condition types into one model, and ControlNet++ [LYK*24] closes a consistency loop on the produced image. Each of these fixes the condition type and, with it, how tightly the output must follow: the operating point is a property of the trained weights. Our cutoff moves that point at inference on one set of weights.

Efficient control on transformer backbones. As backbones moved from UNets to diffusion transformers [PX22, EKB*24], control moved from duplicated encoders to lightweight modules.

ControlNeXt [PWZ*24] removes the parallel branch, OminiControl [TLY*24] reuses the backbone itself as the condition processor, and EasyControl [ZYS*25] and NanoControl [LZL*25] inject conditions through LoRA-style [HSW*21] modules at a small fraction of the parameter cost. Recent work also examines where conditions should enter: Laytrol [HHLZ25] on preserving pretrained knowledge under layout control, PixelControl [LLZ*26] on condition fidelity, and Zhou et al. [ZWC*26] on removing redundant attention in multi-condition DiTs. Our injection-site comparison (Sec. 4.5) reaches a compatible conclusion on a UNet backbone: placement dominates capacity.

Frequency in diffusion. Frequency decomposition appears in diffusion mostly as an intervention on the model’s own signals rather than on its condition. FreeU [SHJL23] rebalances skip and backbone features spectrally at no training cost, FreSca [HLT*25] scales frequency bands of the noise prediction and, like us, uses an energy-based cutoff, DeCo [MWW*25] decouples frequencies in pixel-space diffusion, and FeRA [YHZ*25] routes adaptation by frequency energy. FOCUS [TZY*25] conditions the reverse process on learned frequency priors for test-time adaptation. Closest to us in spirit, Prasain [Pra26] conditions on wavelet bands instead of edge maps for map-to-satellite synthesis. None of them randomises the condition’s *bandwidth* during training so that the band becomes a control at inference. The general pattern — randomise a conditioning parameter while training, expose it as an inference knob — is not itself new: SDXL [PEL*23] samples crop coordinates during data loading and surfaces them as a user control. Our delta is the choice of parameter, and the dissociation that follows from it: bandwidth and adapter strength are not interchangeable knobs.

Structure preservation. StructDiff [HLL*26] targets structure-preserving spatially controllable generation, and sketch-conditioned pipelines have long used edge maps as the structural carrier. The idea that a frequency band is itself an editing axis, and that a drawing should act as a soft constraint rather than a hard one, both predate diffusion in the graphics literature; our motivating sentence borrows from that line and the bibliography below does not yet reflect it. Our condition is a continuum rather than a fixed extractor, which is what makes a dial possible.

Training and evaluation. We train with classifier-free guidance [HS22] dropout and Min-SNR loss weighting [HGL*23], and evaluate with FID [HRU*17] under the clean-FID protocol [PZZ21], LPIPS [ZIE*18] and CLIP score [RKH*21] on FFHQ [KLA18] with BLIP [LLXH22] captions. SDXL and SD3 [PEL*23, EKB*24] are the natural next backbones.

3. Method

Condition. Given an image x , we convert to grayscale, take the 2D FFT, zero every frequency inside a disc of radius $r \cdot \min(H, W)/2$, invert, and min–max normalise to $[0, 1]$. The normalisation is deliberate: it removes the absolute scale of the filter response, so the model has to infer structure density from spatial statistics. We hypothesise that this is what lets a hand drawing, which carries no r , be used at test time; we do not ablate the normalisation.

Condition encoder. A strided convolutional stem reads the condition image directly at 512^2 and reduces it $8\times$. We do not encode the condition with the frozen VAE; Sec. 4.4 shows why.

Injection. Encoder features enter the UNet in two places. Zero-initialised 1×1 heads produce one residual per decoder skip tensor and one for the mid block, added through the UNet’s existing ControlNet interface. In parallel, each cross-attention site attends over a 3×3 window of the condition features, gated by a zero-initialised per-channel vector. Both paths start as exact no-ops: with them switched off the network reproduces stock Stable Diffusion 1.5 [RBL*21] bit for bit, which we assert before every training run.

Training. SD1.5 [RBL*21] stays frozen. We train 6,611,469 parameters on FFHQ-512 [KLA18] with BLIP [LLXH22] captions, split 65,100 train / 1,400 validation / 3,500 test by a seeded shuffle of the 70,000 images (5% test, 2% validation, seed 2026). AdamW, learning rate 10^{-4} , $\beta = (0.9, 0.999)$, cosine schedule with 500 warm-up steps, gradient clipping at 1.0, batch 32, bf16, 12 epochs on one H200. The loss is ϵ -prediction with Min-SNR weighting [HGL*23], $\gamma = 5$. Text conditioning is dropped with probability 0.1 and the structure condition independently with probability 0.05. An EMA of the adapter is kept with a 10-step interval. r is drawn uniformly from $[0, 1]$ per sample. Code and adapter weights accompany the submission.

4. Experiments

4.1. Metrics

We evaluate on 500 of the 3,500 held-out images with a DDIM sampler at 50 steps, CFG 7.5, seed 2026.

Structure consistency. SC converts the generated image to grayscale, applies the *same* high-pass filter at the *same* cutoff the model was given, and takes the Pearson correlation against the given condition over the flattened pixel vector, per image, averaged over images. Both signals are the per-image min–max normalised filter output, so the correlation is scale free. Two anchors bound it: feeding the ground truth back gives exactly 1.000 at every r , and an unrelated face gives 0.122 at $r=0$ and ≈ -0.02 above it. Because the instrument moves with r , a row-to-row difference mixes model behaviour with instrument change, so we report a cutoff-independent metric beside it.

Edge-F1. Canny edges (thresholds 100 and 200 on the 8-bit grayscale image) are taken from the generation and from the reference and matched with a 2-pixel tolerance, implemented as a 5×5 dilation of each map before intersecting; F1 is computed per image and averaged. It does not depend on r , which is why it appears in Table 1: 0.7796 at $r=0.05$ falling to 0.4724 at $r=1$, so the dial survives a cutoff-independent instrument, with a floor near 0.4724 that aligned FFHQ faces supply for free.

Diversity. Mean pairwise LPIPS (AlexNet backbone) over $K=4$ samples generated from one condition with different seeds; higher means the condition constrains less. It is not independent of

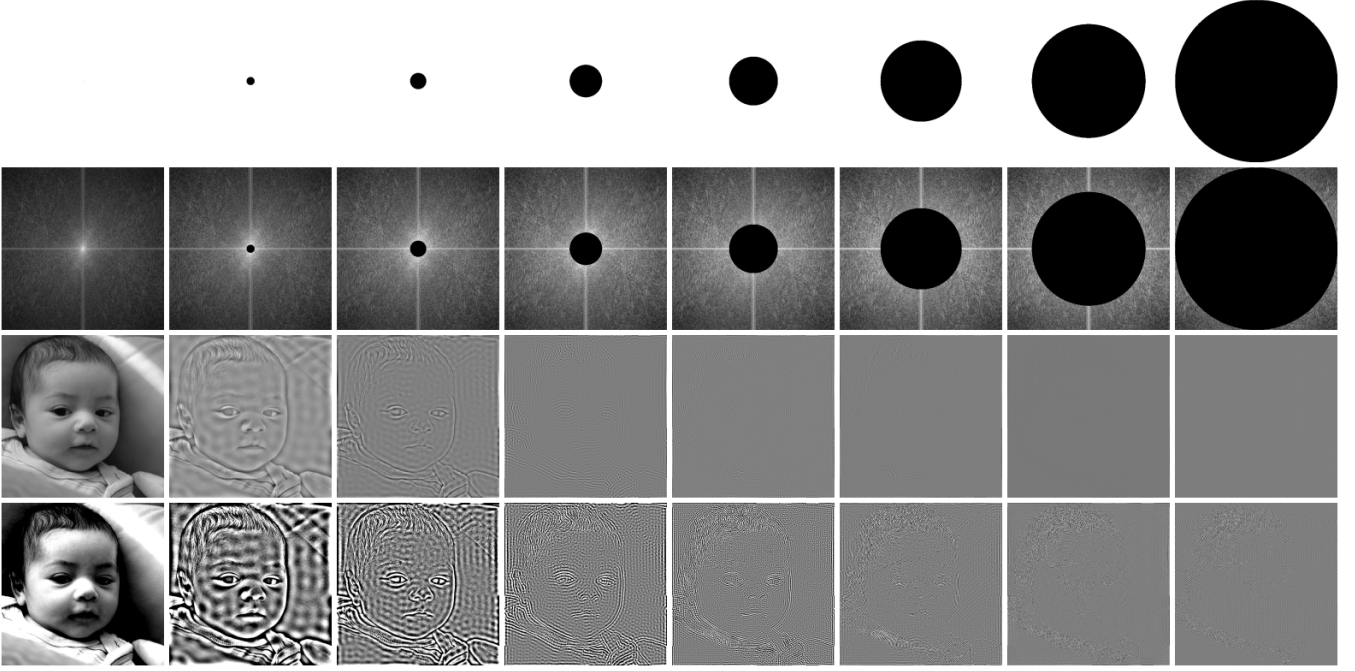


Figure 1: The dial, from the frequency side. Columns are $r = 0, 0.05, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0$. **Row 1:** the radial mask; black is removed. **Row 2:** the log-magnitude spectrum after masking, with the hole growing outward. **Row 3:** the resulting condition, exactly as the model receives it. **Row 4:** the same data displayed at $\pm 2\sigma$ so that the residual content is visible. Above $r \approx 0.2$ the condition looks like flat grey — its standard deviation falls from 0.196 at $r=0$ to 0.041 at $r=1$ — yet the face outline survives in row 4, and the model still reaches SC 0.4019 at $r=0.2$. Low contrast is not the same as no information, which is what the per-image normalisation forces the model to learn.

the LPIPS column: across the sweep LPIPS — Diversity is 0.0320 ± 0.0038 , nearly constant over a range of 0.20, so the two carry close to the same information and we draw no separate conclusion from Diversity alone.

FID and CLIP. Clean-FID [HRU*17, PZZ21] between the 500 generations and the same 500 references, and CLIP score with ViT-B/32 [RKH*21] against each image’s own caption.

4.2. The dial

Table 1 gives the operating curve. On $r \in [0.05, 0.3]$ structure consistency falls at every step and sample diversity rises at every step, which is the trade-off the dial is meant to expose. Two qualifications belong with that sentence. Going from $r=0$ to $r=0.05$ structure consistency rises by 0.0479 ($z = 6.24$, paired within one checkpoint), so the sweep is not monotone at its left end; the same rise appears in Edge-F1 (0.7776 to 0.7796), so it is model behaviour rather than an artefact of the SC instrument. And at the right end $r=0.5$ and $r=0.7$ are not separable across seeds (paired $t = 0.732$, $df = 2$, critical 4.303). FID stays between 50.13 and 57.40 across the entire sweep.

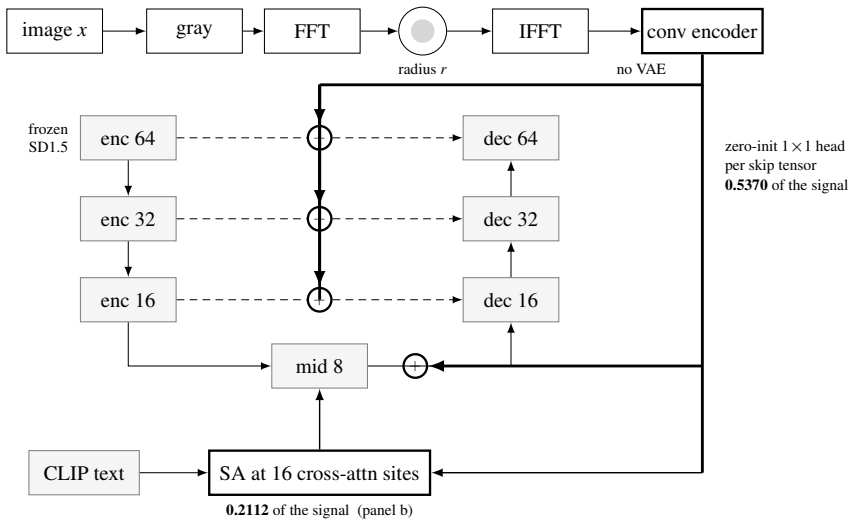
Figure 3 holds the seed and the caption fixed and varies only r . Figure 4 plots the curve and, below it, both dials in the same structure-consistency/FID plane.

r	SC $\pm$ SEM	seed sd	Edge-F1	Diversity	FID
0.00	0.6991 ± 0.0070	0.0105	0.7776	0.2994	50.13
0.05	0.7469 ± 0.0030	0.0015	0.7796	0.3126	51.58
0.10	0.6211 ± 0.0035	0.0059	0.7439	0.3414	54.68
0.20	0.4019 ± 0.0037	0.0098	0.7113	0.3719	53.45
0.30	0.2347 ± 0.0041	0.0106	0.6878	0.3794	52.77
<i>beyond the range we claim</i>					
0.50	0.1259 ± 0.0047	0.0093	0.6225	0.4100	53.97
0.70	0.1110 ± 0.0047	0.0186	0.5788	0.4334	53.69
1.00	0.0010 ± 0.0006	0.0002	0.4724	0.5066	57.40

Table 1: Operating curve, 500 of the 3,500 test images. SEM is over those 500 images; seed sd is over three training seeds. Edge-F1 is a cutoff-independent reading of the same runs and is reported alongside SC because SC re-extracts at the same cutoff the model was given (Sec. 4.1). FID has a single estimate per cell and no variance; we use it for row-to-row comparison only.

Across seeds 2026, 2027 and 2028 the curve is stable: SC at $r=0.05$ is 0.7479 ± 0.0015 and at $r=0.3$ it is 0.2318 ± 0.0106 .

(a) condition path and the two injection sites



(b) inside one site

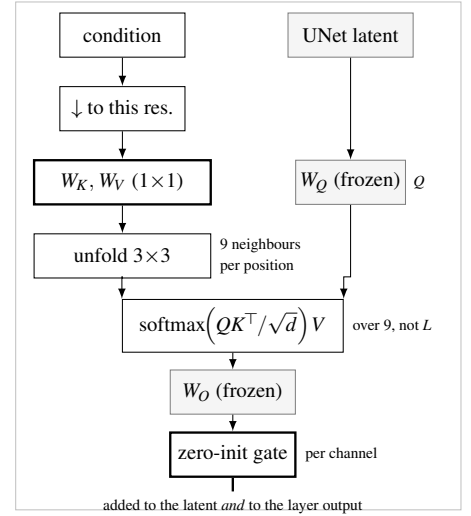


Figure 2: (a) A disc of radius r is masked out of the spectrum and a trainable convolutional encoder reads the result at full resolution, never through the VAE. Its features enter the frozen UNet twice: as zero-initialised residuals on every skip tensor ($\oplus$), and through StandAlone attention at the 16 cross-attention sites. Stroke weight follows the measured split — the skip path carries 0.5370 of the structure signal against 0.2112 for attention (Sec. 4.5). (b) One StandAlone site. The query comes from the UNet’s own latent through the frozen W_Q ; keys and values come from the condition through trained 1×1 projections, unfolded so each position attends over its 3×3 neighbourhood rather than the whole map. The result passes the frozen output projection and a zero-initialised per-channel gate, then is added both into the latent — so the text query below sees structure — and to what the layer returns.

Setting	SC	Diversity	FID
scale 0.25	0.0408	0.7752	104.95
scale 0.50	0.2146	0.6061	76.46
scale 0.75	0.4893	0.4171	58.90
scale 1.00	0.6211	0.3414	54.68
$r=0.3$ (ours)	0.2347	0.3794	52.77
$r=0.5$ (ours)	0.1259	0.4100	53.97

Table 2: Strength dial against frequency dial. Reading across matched SC: scale 0.50 reaches 0.2146 at FID 76.46, while $r=0.3$ reaches 0.2347 at FID 52.77.

4.3. Strength is not the same knob

An obvious alternative is to train on one cutoff and scale the adapter’s contribution at inference. We tested it on the same checkpoint, holding $r=0.1$ and scaling both injection paths.

Weakening the adapter degrades the image; narrowing the condition’s bandwidth does not. The frequency parameterisation is what makes the trade-off traversable.

4.4. The VAE eats the condition

Encoding the condition with the frozen SD VAE loses most of it. An image-prompt design such as IP-Adapter [YZL*23] takes a different route — a CLIP image encoder and a projection — so this measurement does not speak to it. Figure 5 plots it. Passing the con-

dition through encode–decode and re-measuring leaves 0.8562 at $r=0.1$, 0.5773 at $r=0.3$ and 0.0700 at $r=0.7$. The round trip lower-bounds what the latent retains — a component may survive there that the frozen decoder cannot render — but an adapter reading that latent cannot recover what the round trip has lost. Building the condition at 256 and upsampling moves the structure to lower absolute frequency and preserves far more (0.7978 at $r=0.3$), which is a data-pipeline artefact worth knowing about when comparing against older setups.

4.5. Where to inject

Table 3 lists absolute configurations; each row below the control differs from the *conv / attn* row in one field. Skip-connection residuals dominate. Figure 6 decomposes them. Ablating them away and keeping only cross-attention drops SC at $r=0$ from 0.6991 to 0.2112; keeping only the skip path retains 0.5370. With both paths off the model collapses to 0.0368 SC and FID 120.50, which is the control that says the adapter is doing the work. The split is not the same on both axes: on SC the skip path leads by roughly $2.5\times$, but on FID the two paths are almost indistinguishable (65.12 against 67.07), so the placement effect we measure is a structure-fidelity effect and not a general image-quality one.

Capacity helps much less: widening the projection to a 128-channel stem raises SC at $r=0.05$ from 0.1485 to 0.2826, and doubling again to 256 reaches only 0.2986. A 7×7 attention window is worse than 3×3 (0.0359). All of these vary capacity with the



Figure 3: One seed, one caption, r swept. Columns: reference, then $r = 0, 0.05, 0.1, 0.3, 0.5, 0.7, 1.0$. The background crowd thins out, the eyewear drifts from the reference frame, and the clothing colour detaches, while identity and pose survive. The reference is never shown to the model.

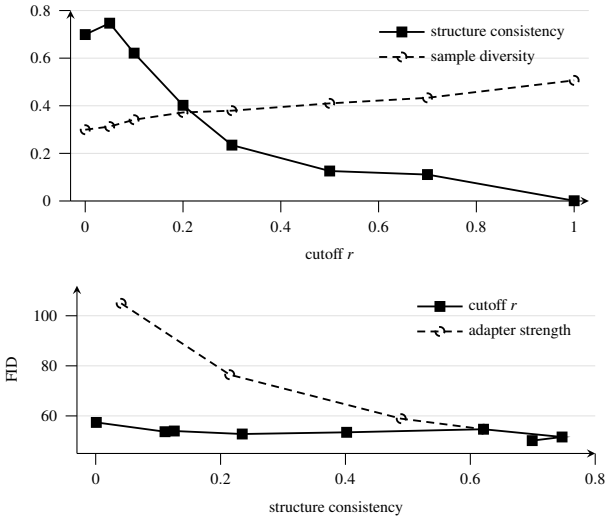


Figure 4: *Top:* structure consistency falls and sample diversity rises as the cutoff grows. *Bottom:* the same runs in the structure-consistency / FID plane. Narrowing the condition’s bandwidth traverses the axis at nearly constant FID; weakening the adapter buys the same structure consistency by degrading the image. Both panels are plotted directly from the evaluation outputs.

skip path absent, so they bound how far capacity carries *that* design rather than settling capacity against placement in general.

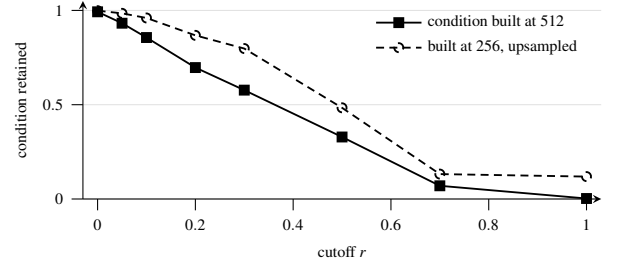


Figure 5: Correlation between the condition fed to the autoencoder and the same condition re-extracted after an encode–decode round trip, 24 images. Above $r \approx 0.5$ the condition is effectively gone, so an adapter reading the VAE latent cannot recover it. Building the condition at 256 and upsampling puts the structure at lower absolute frequency and preserves far more, which is why comparisons against older pipelines need the source resolution stated.

4.6. Against per-cutoff specialists

We trained four adapters at fixed r and compared each against the single randomised- r model at its own cutoff. Because the single model was trained three times, we express each gap against its three-seed spread rather than against a within-run standard error; a gap smaller than the two-sided t critical value at $df = 2$ (4.303) is not separable. The result is mixed: the specialist wins at $r=0.1$ (0.7049 against 0.6245, $\Delta = +0.0803$, +13.55 sd) and the single model wins at $r=0.3$ (0.2318 against 0.1214, -0.1104 , -10.42 sd). Neither $r=0.5$ (-1.00 sd) nor $r=0.7$ ($+0.14$ sd) separates. Our earlier single-seed reading called $r=0.7$ a marginal win for the specialist; across seeds it is a tie.

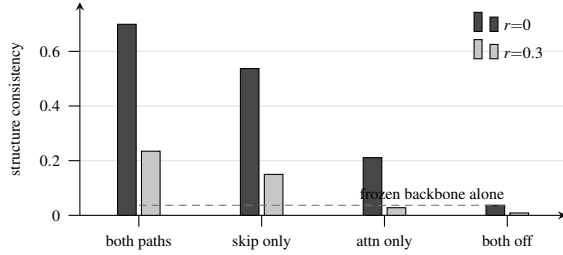


Figure 6: Both injection paths on, then each alone, then neither. The skip residuals carry most of the structure signal; cross-attention alone recovers little of it, and with both off the model falls to the frozen backbone (FID 120.50 against 50.13 with both paths on). The two paths are close to additive.

Cond. encoder	Injection	SC $r=0.05$	SC $r=0.1$	FID $r=0.1$
no adapter (control)		0.0368	0.0082	120.50
frozen VAE	attn, no gate	0.0234	0.0147	76.25
frozen VAE	attn	0.0303	0.0177	65.71
conv	attn	0.1485	0.0772	56.39
conv, 128	attn	0.2826	0.1882	53.17
conv, 256	attn	0.2986	0.2158	52.84
conv	attn, 7×7	0.0359	0.0218	58.51
conv	attn + skip	0.7469	0.6211	54.68

Table 3: Absolute configurations, not a cumulative chain: every row below the control differs from the conv / attn row in exactly one field, and all rows use a 3×3 window and a zero-initialised gate unless stated. The main model is the last row and carries no stem, i.e. the minimal 1×1 projection. The first three rows sit at or below the no-adapter control, so at that level a weak design and a run that failed to learn are not distinguishable. Capacity is varied only in the attention-only regime; no row varies stem width with skips present, so the table supports a statement about these two injection designs rather than a general ordering of placement over capacity.

One model therefore does not dominate four. What it does instead is give a dial the specialists cannot: a model trained at $r=0.1$ has never seen $r=0.3$, so its curve is a single point.

4.7. Baselines

Against pretrained ControlNet [ZRA23] on its own condition, over the same test images and captions: ControlNet-Canny reaches Edge-F1 0.6854, LPIPS 0.5294, FID 71.92 with 361,279,120 trainable parameters; ControlNet-HED reaches 0.5733 / 0.6548 / 118.98. At $r=0.1$ our adapter reaches Edge-F1 0.7439, LPIPS 0.3748 and FID 54.68 with 6,611,469 parameters. Two asymmetries make this a comparison of operating points rather than a like-for-like control: the conditions differ (Canny edges against a frequency band), and our adapter was trained on 65,100 FFHQ images while ControlNet is zero-shot here and FID is measured against FFHQ statistics. On text alignment we do not win anywhere: our best CLIP over the whole sweep is 30.30, below ControlNet-Canny’s 30.34 (ControlNet-HED reaches 29.16). Training Control-



Figure 7: Hand-drawn face sketches (odd columns) and what the model makes of them (even columns), zero-shot. The drawings carry no cutoff; the model has to read structure density off the drawing itself, which the per-image min-max normalisation permits. Most of the scans carry the archive filename stamped across the top, and the model redraws it — a literal reading of the condition it was given. Removing the whole top 20% of the frame (102 rows), a generous over-estimate of that band, moves the score to 0.5398 (-0.0086 , 0.32 standard errors), so the band is not what the score is measuring.

Net on our condition and data would be the matched control; we have not run it.

4.8. Hand-drawn sketches

Figure 7 shows the transfer. On 40 real face drawings the model was never trained on, structure consistency against the input drawing is 0.5484 and CLIP score is 23.43. FID is not reported: 40 images is too few for a stable estimate, and our evaluator refuses rather than print one. We also ran 500 general-scene drawings, far outside the training domain, where structure consistency falls to 0.1714 and FID against FFHQ is 285.91: the model follows the drawing but does not produce a face. That is the expected failure for a face model and we report it rather than omit it.

5. Limitations

The useful range is roughly $r \in [0, 0.3]$. Above it the condition retains little energy — 0.0014 of the spectrum at $r=0.5$ — and structure consistency becomes hard to interpret, since two nearly empty signals correlate near zero whatever the model does.

Everything here is one domain (FFHQ faces) and one backbone (SD1.5). Current structure-control work has moved to DiT backbones [PX22, EKB*24], where LoRA-style [HSW*21] adapters [TLY*24, ZYS*25, LZL*25] report control at a fraction of our parameter count; our efficiency numbers should be read against ControlNet, not against that line.

The specialist comparison is honest but unflattering at $r=0.1$, and we have not run a user study.

6. Conclusion

Randomising the bandwidth of a structure condition during training turns the cutoff into an inference-time control, and that control is not reproducible by scaling the adapter down. Getting there required fixing where the condition enters the network and keeping it out of the frozen autoencoder; both were measured on the SD1.5 autoencoder and UNet, and we expect them to carry to conditioning designs built on that family. We have not tested others.

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